The normal distribution

1: Characteristics of the Normal Distribution

What shape is it? What parameters does it have?

2: Finding probabilities on a standard normal curve.

"Given that IQ is distributed as $X \sim N\left(100, 15^2\right)$, determine the probability that a randomly chosen person has an IQ above 130."

3: Finding unknown means/standard deviations.

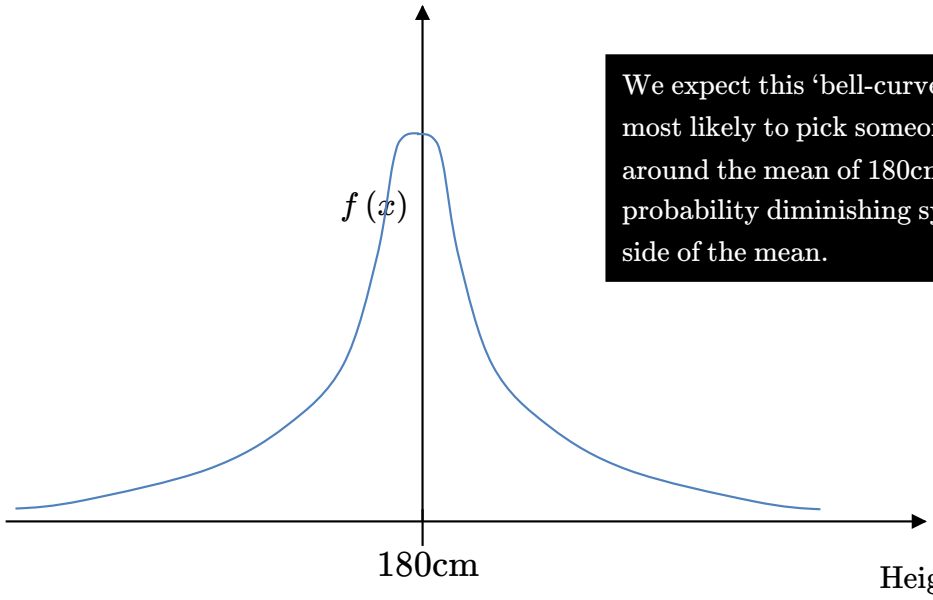
In Wales, 30% of people have a height above 1.6m. Given the mean height is 1.4m and heights are normally distributed, determine the standard deviation of heights.

4: Binomial $\rightarrow$ Normal Approximations

How would I approximate $X \sim B(10, 0.4)$ using a Normal distribution? Under what conditions can we make such an approximation?

5: Hypothesis Testing

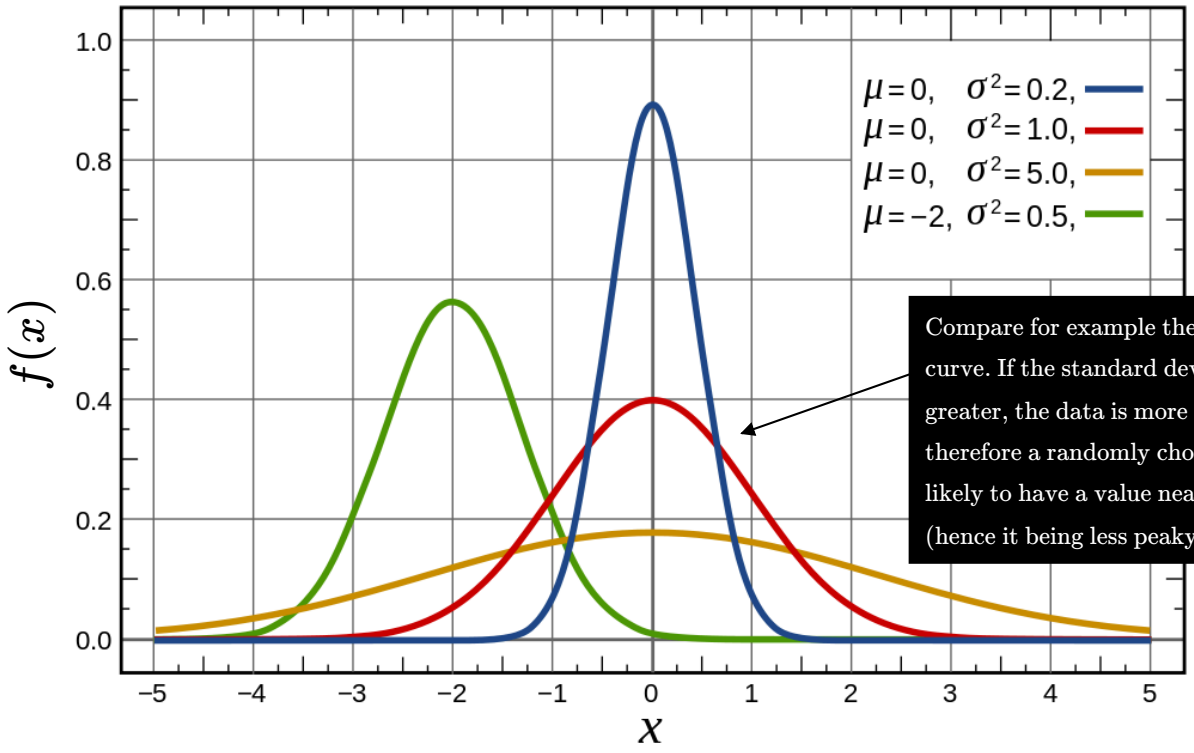
The following shows what the probability distribution might look like for a random variable X, if X is the height of a randomly chosen person.



We expect this 'bell-curve' shape, where we're most likely to pick someone with a height around the mean of 180cm, with the probability diminishing symmetrically either side of the mean.

A variable with this kind of distribution is said to have a **normal distribution**.

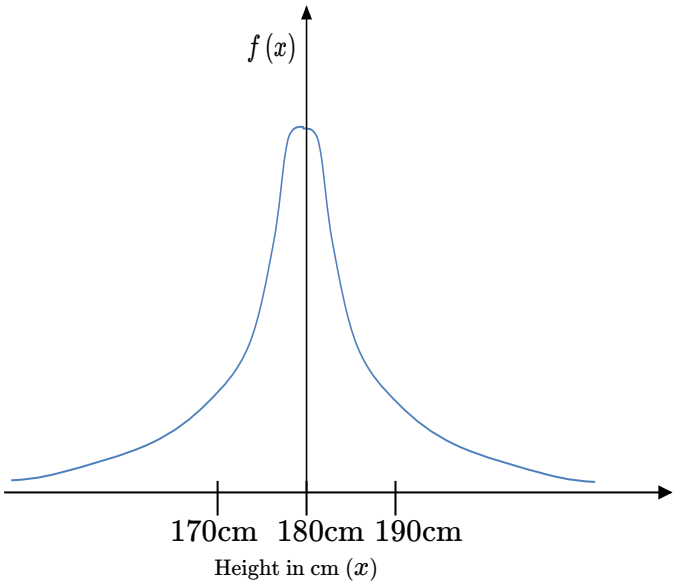
For normal distributions we tend to draw the y axis at the mean for symmetry.



Compare for example the blue and red curve. If the standard deviation is greater, the data is more spread out, and therefore a randomly chosen thing is less likely to have a value near the mean (hence it being less peaky).

We can set the mean μ and the standard deviation σ of the Normal Distribution. If a random variable X is normally distributed, then we write

$$\mathbf{X \sim N\left(\mu, \sigma^2\right)}$$



Q1. What kind of variable can be modelled by a normal distribution?

Q1

Q2. Discrete probabilities sum to 1. What condition holds for a continuous distribution?

Q2

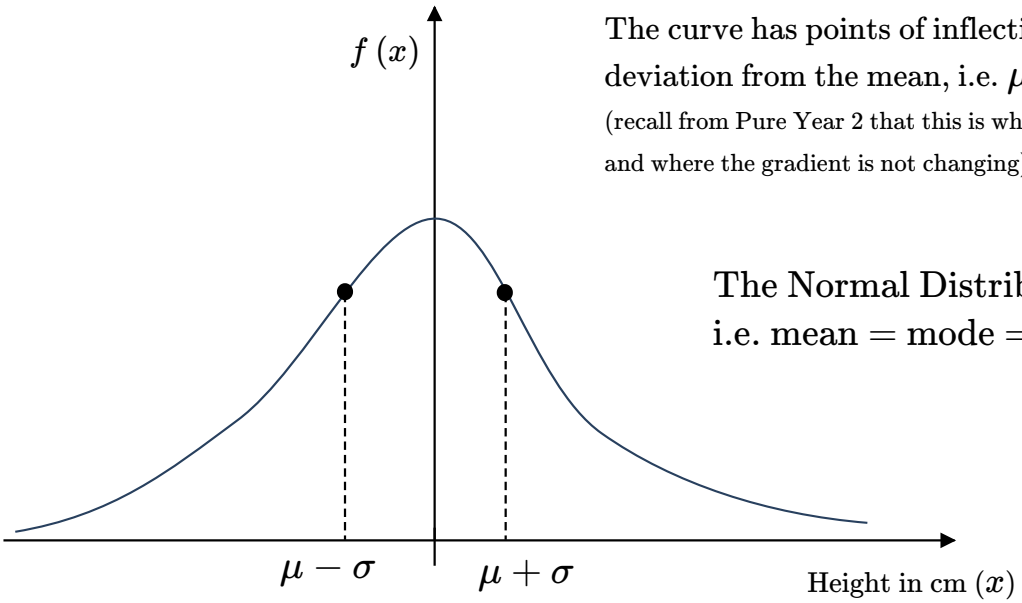
Q3. How do we find $P(170 < X < 190)$?

Q3

Q4. What is the probability that height is exactly 200 cm?

Q4

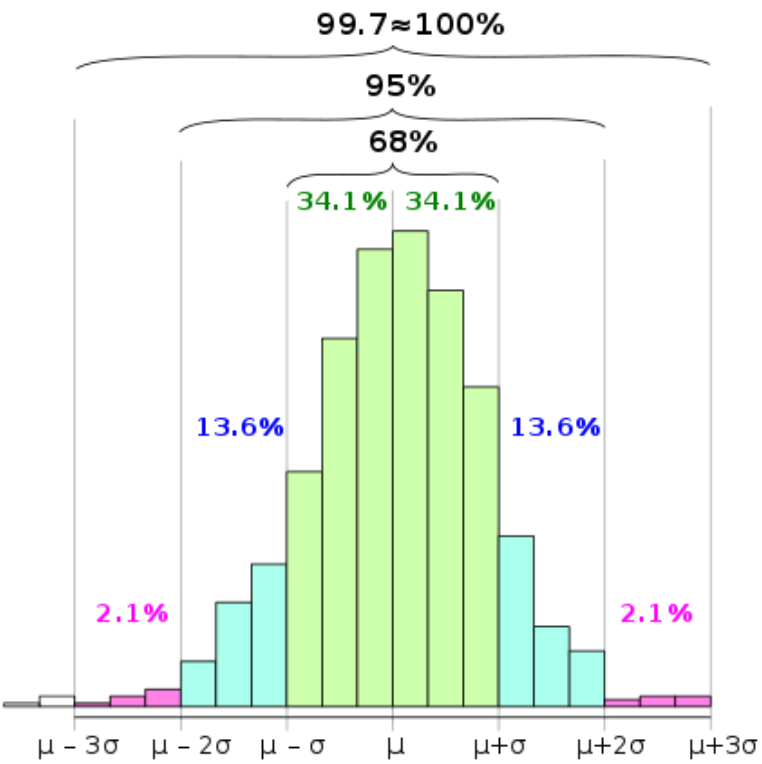
The vertical axis gives probability density per centimetre. As in a histogram, area represents probability. A continuous variable has a density function $f(x)$.



The curve has points of inflection one standard deviation from the mean, i.e. $\mu \pm \sigma$

(recall from Pure Year 2 that this is where the curve changes concavity, and where the gradient is not changing)

The Normal Distribution is symmetrical, i.e. mean = mode = median



The histogram above is for a quantity which is approximately normally distributed.

The 68-95-99.7 rule is a shorthand used to remember the percentage of data that is within 1, 2 and 3 standard deviations from the mean respectively.
You need to memorise this!

$\approx 68\%$ of data is within one standard deviation of the mean.
$\approx 95\%$ of data is within two standard deviations of the mean.
$\approx 99.7\%$ of data is within three standard deviations of the mean.

For practical purposes we consider all data to lie within $\mu \pm 5\sigma$

[Textbook] The diameters of a rivet produced by a particular machine, X mm, is modelled as $X \sim N\left(8, 0.2^2\right)$. Find:
a) $P\left(X > 8\right)$
b) $P\left(7.8 < X < 8.2\right)$

Tip: Draw a diagram!

a

b

IQ ("Intelligence Quotient") for a given population is, by definition, distributed using $X \sim N\left(100, 15^2\right)$. Find:

a) $P\left(70 < X < 130\right)$

b) $P\left(X > 115\right)$

Tip: Draw a diagram!

a

b

6. Dormouse masses D grams follow $N(\mu, 25)$. If 97.5% weigh less than 70 g, find μ.

7. Pig masses M kg follow $N(\mu, \sigma^2)$. If 84% weigh more than 52 kg and 97.5% weigh more than 47.5 kg, find μ and σ^2.

a

b

8. Test scores S (percentages) follow a normal model with mean 45 and standard deviation 15. Find:

(a) $P(S > 45)$

(b) $P(30 < S < 60)$

(c) $P(15 < S < 75)$

(d) Alexia says this model is unsuitable since scores above 100% are impossible. Is she correct? Give a reason.

a

b

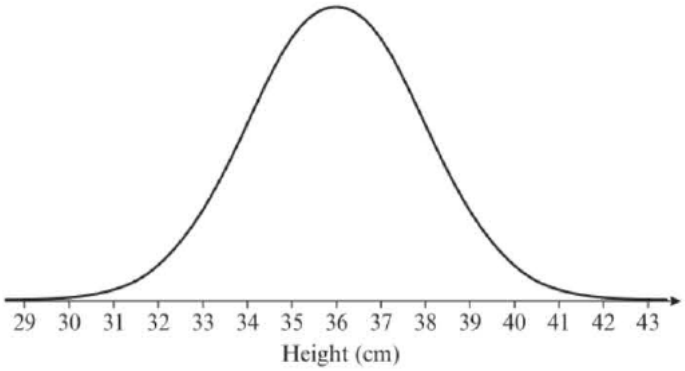
c

d

9. The diagram shows UK barn owl heights in cm. An ornithologist models them using a normal distribution.

(a) State the mean height.

(b) Estimate the standard deviation.



a

b

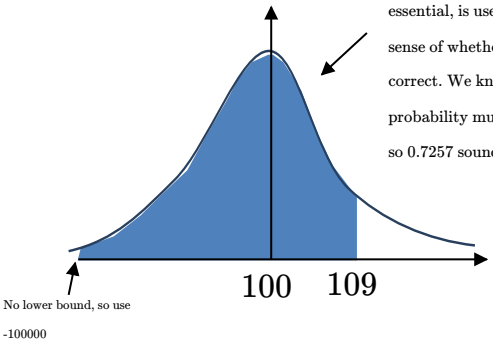
https://education.casio.co.uk/resources/videos/normal-distribution-probabilities-and-values

Just like a cumulative frequency graph gives the running total of the frequency up to a given value, a **cumulative distribution** gives the **running total of the probability** up to a given value.

IQ is distributed using $X \sim N\left(100, 15^2\right)$. Find
(a) $P\left(X < 109\right)$
(b) $P\left(X \geq 93\right)$
(c) $P\left(110 < X < 120\right)$
(d) $P\left(X < 80 or X > 106\right)$

a

1. Press MODE. Stats
2. Choose DISTRIBUTION
3. Choose Normal CD (i.e. "Cumulative Distribution")
4. Since the lower value is effectively $-\infty$, use any value at least 5σ below the mean (-100000 will do!). Press = after each value.
5. Put the upper value as 109.
6. Set $\sigma = 15$ and $\mu = 100$
7. You should obtain $P\left(X < 109\right) = 0.7257(4dp)$



A diagram, while not essential, is useful to get a sense of whether our answer is correct. We know the probability must be above 0.5, so 0.7257 sounds about right.

IQ is modelled by $X \sim N(100, 15^2)$. Find:
(a) $P(X < 109)$
(b) $P(X \geq 93)$
(c) $P(110 < X < 120)$
(d) $P(X < 80 \text{ or } X > 106)$.

b b

Note: $\geq$ vs $>$ makes no difference as the distribution is continuous.

c c

d d

Tip. Return to normal calculation mode to subtract from 1. ANS stores the last distribution result.

Test Your Understanding — Luisa · 15

Assume IQ $X \sim N(100, 15^2)$. Mensa requires an IQ of at least 131.

(a) What percentage of the population is eligible?

(b) Of 30 randomly chosen adults, what is the probability that at least three are eligible? Hint: use the binomial distribution.

a

b

Exercise 3B – Statistics Y2 pages 40-41 — Luisa · 16

8. Soap volume per press, X ml, follows $N(6, 0.8^2)$.

(a) Find (i) $P(X > 7)$ and (ii) $P(X < 5)$.

(b) Find the probability that all three presses dispense less than 5 ml.

8a(i)

8a(ii)

8b

9. Bottled mineral water volume W ml follows $N(500, 14^2)$.

(a) Find (i) $P(W > 505)$ and (ii) $P(W < 490)$.

(b) Find the probability all four sampled bottles contain more than 490 ml.

9a(i)

9a(ii)

9b

11. Bolt diameters D mm follow $N(13, 0.1^2)$.

(a) Find the probability a bolt is smaller than 12.8 mm. (1)

(b) A perfect bolt has diameter between 12.9 and 13.1 mm. In a random sample of 40, find the probability of more than 25 perfect bolts. (4)

a

b

12. Squirrel masses X g follow $N(480, 40^2)$.

(a) Find the probability a squirrel has mass above 490 g. (1)

(b) A random sample of 30 is taken. Find the probability at least 15 have masses between 470 g and 490 g. (4)

a

b

Inverse Normal Distribution

Luisa · 19

We now know how to use a calculator to value of the variable to obtain a probability. But we might want to do the reverse: given a probability of being in a region, how do we find the value of the boundary?

[Textbook] $X \sim N\left(20, 3^2\right)$. Find, correct to two decimal places, the values of a such that:

a. $P(X < a) = 0.75$

b. $P(X > a) = 0.4$

c. $P(16 < X < a) = 0.3$

a

b

c

Further Example

Luisa · 20

If the IQ of a population is distributed using $X \sim N\left(100, 15^2\right)$.

a. Determine the IQ corresponding to the top 30% of the population.

b. Determine the interquartile range of IQs.

a

b

$X \sim N\left(80, 7^2\right)$. Using your calculator,
a. determine the a such that $P(X > a) = 0.65$
b. determine the b such that $P(75 < X < b) = 0.4$
c. determine the c such that $P(c < X < 76) = 0.2$
d. determine the interquartile range of X.

a

c

b

d

6. Badger masses M kg follow $N(4.5, 0.6^2)$.
(a) Find the lower quartile.
(b) Find the 80th percentile.
(c) Explain without calculation why the median is 4.5 kg.

a

b

c

7. Learner drivers' theory scores X follow $N(72, 6^2)$.

(a) Find a such that $P(X < a) = 0.6$. (1)

(b) Find the interquartile range. (2)

7a

7b

8. Chocolate bar masses Y g follow $N(60, 2^2)$.

(a) Find y such that $P(Y > y) = 0.2$. (1)

(b) Find the 10% to 90% interpercentile range. (2)

(c) Tom says the median equals the mean. Is he correct? Explain. (1)

8a

8b

8c

9. Men's heights H cm follow $N(170, 10^2)$. A manufacturer makes frock coats, which reach from neck to near floor.

Short	Regular	Long
30%	50%	20%

(a) Suggest a height range to advertise for regular coats. (4)

(b) State one assumption. (1)

a

b

Z is the number of standard deviations above the mean.

Using the above definition, and considering the a variable (IQ) distributed as $X \sim N\left(100, 15^2\right)$,calculate (in your head!):

IQ	Z
100	
130	
85	
165	
62.5	

What calculation are you actually doing in your head?

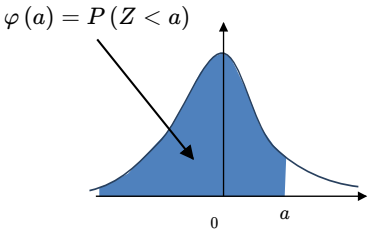
For an IQ of 130: $Z = \frac{130-100}{15} = 2$ as expected.

! Z represents the coding:
$Z = \frac{X-\mu}{\sigma}$
and $Z \sim N\left(0, 1^2\right)$. Z is known as a **standard** normal distribution.

The 0 and 1 of $Z \sim N\left(0, 1^2\right)$ are the result of the coding. If we've subtracted μ from each value the mean of the normal distribution is now 0. If we've divided all the values by σ the standard deviation is now $\frac{\sigma}{\sigma} = 1$

The point of coding in this context is that all different possible normal distributions become a single unified distribution where we no longer have to worry about the mean and standard deviation.
It means for example when we calculate $P\left(Z < 3\right)$, this will always give the same probability regardless of the original distribution.

It also means we can look up probabilities in a **z-table**:



$\Phi\left(a\right) = P\left(Z < a\right)$ is the cumulative distribution for the standard normal distribution. The values of $\Phi(a)$ can be found in a z-table.

This is a traditional z-table in the old A Level syllabus (but also found elsewhere). You no longer get given this and are expected to use your calculator.

z	$\Phi(z)$	z	$\Phi(z)$	z	$\Phi(z)$	z	$\Phi(z)$	z	$\Phi(z)$
0.00	0.5000	0.50	0.6915	1.00	0.8413	1.50	0.9332	2.00	0.9772
0.01	0.5040	0.51	0.6950	1.01	0.8438	1.51	0.9345	2.01	0.9778
0.02	0.5080	0.52	0.6985	1.02	0.8461	1.52	0.9357	2.02	0.9783
0.03	0.5120	0.53	0.7019	1.03	0.8485	1.53	0.9370	2.03	0.9788
0.04	0.5160	0.54	0.7054	1.04	0.8508	1.54	0.9382	2.04	0.9793
0.05	0.5199	0.55	0.7088	1.05	0.8531	1.55	0.9394	2.05	0.9798
0.06	0.5239	0.56	0.7123	1.06	0.8554	1.56	0.9406	2.06	0.9803
0.07	0.5279	0.57	0.7157	1.07	0.8577	1.57	0.9418	2.07	0.9808
0.08	0.5319	0.58	0.7190	1.08	0.8599	1.58	0.9429	2.08	0.9812
0.09	0.5359	0.59	0.7224	1.09	0.8621	1.59	0.9441	2.09	0.9817
0.10	0.5398	0.60	0.7257	1.10	0.8643	1.60	0.9452	2.10	0.9821

This is from the new formula booklet. This is sometimes known as a 'reverse z-table', because you're looking up the z-value for a probability. Beware: p here it the probability of **exceeding** z rather than being up to z. Let's use it...

p	z	p	z
0.5000	0.0000	0.0500	1.6449
0.4000	0.2533	0.0250	1.9600
0.3000	0.5244	0.0100	2.3263
0.2000	0.8416	0.0050	2.5758
0.1500	1.0364	0.0010	3.0902
0.1000	1.2816	0.0005	3.2905

Examples — Luisa · 27

$X \sim N(50, 4^2)$. Express (a) $P(X < 53)$ and (b) $P(X \geq 55)$ in terms of $\Phi(z)$.

a

b

Standardise to turn X into Z.

Adult systolic blood pressure S mmHg is normal, with mean 127 and standard deviation 16. A researcher studies adults above the 95th percentile. Find the minimum blood pressure for inclusion.

Blood pressure

p	z	p	z
0.5000	0.0000	0.0500	1.6449
0.4000	0.2533	0.0250	1.9600
0.3000	0.5244	0.0100	2.3263
0.2000	0.8416	0.0050	2.5758
0.1500	1.0364	0.0010	3.0902
0.1000	1.2816	0.0005	3.2905

A sketch helps determine whether z is positive or negative. Once the table gives z, use $z = (s - \mu)/\sigma$.

Further Examples — Luisa · 28

Use the standard normal distribution to answer each question.

p	z	p	z
0.5000	0.0000	0.0500	1.6449
0.4000	0.2533	0.0250	1.9600
0.3000	0.5244	0.0100	2.3263
0.2000	0.8416	0.0050	2.5758
0.1500	1.0364	0.0010	3.0902
0.1000	1.2816	0.0005	3.2905

(a) Find $P(Z > -1.3)$.

a

(b) Find $P(-2 < Z < 1)$.

b

(c) Find a if $P(Z > a) = 0.7$.

c

(d) Find a if $P(-a < Z < a) = 0.6$.

d

Test Your Understanding

1. IQ has mean 100 and standard deviation 15. Use a normal table to find the IQ cutoff for (a) the top 10% and (b) the bottom 20%.

2. $X \sim N(100, 15^2)$. Express in terms of Φ:
(a) $P(X > 115)$
(b) $P(77.5 < X < 112)$.

3. Find a such that:
(a) $P(-a < Z < a) = 0.2$
(b) $P(0 < Z < a) = 0.35$.

p	z	p	z
0.5000	0.0000	0.0500	1.6449
0.4000	0.2533	0.0250	1.9600
0.3000	0.5244	0.0100	2.3263
0.2000	0.8416	0.0050	2.5758
0.1500	1.0364	0.0010	3.0902
0.1000	1.2816	0.0005	3.2905

1a

2a

3a

1b

2b

3b

Exercise 3D – Statistics Y2 pages 49

5. (a) Use a percentage points table to find z such that $P(Z > z) = 0.025$. (1)
(b) A fighter jet training programme takes the top 2.5% of candidates. Test scores are normal with mean 80 and standard deviation 4. Find the necessary score. (2)

a

b

6. (a) Use a percentage points table to find z such that $P(Z < z) = 0.15$. (1)

(b) A petite hat should fit the smallest 15% of customers. Hat sizes are normal with mean 57 cm and standard deviation 2 cm. Find the hat size. (2)

a

b

7. (a) Use a percentage points table to find the z-values for the 10% to 90% interpercentile range. (2)

(b) Bulb life is normal with mean 1175 hours and standard deviation 56 hours. A standard bulb's life lies in that range. Find the endpoints to the nearest hour. (2)

a

b

Finding the missing mean or standard deviation

In the last section, you may have thought, "what's the point of standardising to Z when I can just use the DISTRIBUTION mode on my calculator?"

HINT: both forward and reverse normal lookups on the calculator **required you to specify μ and σ.**

[Textbook] $X \sim N\left(\mu, 3^2\right)$. Given that $P\left(X > 20\right) = 0.2$, find the value of μ.

In the last section, you may have thought, "what's the point of standardising to Z when I can just use the DISTRIBUTION mode on my calculator?"

[Textbook] A machine makes metal sheets with width, X cm, modelled as a normal distribution such that $X \sim N\left(50, \sigma^2\right)$.
(a) Given that $P\left(X < 46\right) = 0.2119$, find the value of σ.
(b) Find the 90th percentile of the widths.

The method here is exactly the same as before:

1. Using a sketch, determine whether you're expecting a positive or negative z value.
2. Look up z value, using tables if you can (otherwise your calculator). Make negative if in bottom half.
3. Use $Z = \frac{X-\mu}{\sigma}$

When both are missing

Edexcel S1 Jan 2011. Machine B fills cartons with soup. The mass Y grams is normal with mean μ and standard deviation σ.
(c) Given $P(Y < 160) = 0.99$ and $P(Y > 152) = 0.90$, find μ and σ. (6)

(c)

Test Your Understanding — Luisa · 35

Edexcel S1 May 2013(R). London–Berlin flight times are normal with mean 100 minutes and standard deviation d.

(b) 15% take more than 115 minutes. Find d. (4)

b

Edexcel S1 Jan 2002 Q5. Pregnancy durations for a breed of cow are normal. 2.5% are shorter than 235 days and 15% exceed 286 days.

(a) Show $\mu - 235 = 1.96\sigma$. (2)

(b) Obtain a second equation. (3)

(c) Find μ, σ. (4)

(d) Find the limits containing the middle 68.3%. (2)

5a

5b

5c

5d

Exercise 3E – Statistics Y2 pages 51-53 — Luisa · 36

11. Pottery bowl diameters D mm are normal with standard deviation 5 mm. 75% exceed 200 mm.

(a) Find the mean μ. (2)

(b) Find $P(204 < D < 206)$. (1)

(c) Three bowls are selected at random. Find the probability all exceed 205 mm. (3)

a

b

c

12. Cloth thickness T mm is normal with mean 2.5 mm. 65% of cloths are thinner than 2.55 mm.
(a) Find the standard deviation. (2)
(b) Find the proportion between 2.4 mm and 2.6 mm thick. (1)
(c) Only cloths in this range can be sold. Find the probability at least 15 of a random sample of 20 can be sold. (3)

a

b

c

13. Penguin masses M kg are normal with mean μ and standard deviation σ. 10% weigh under 18 kg and 5% over 30 kg.
(a) Sketch this information. (2)
(b) Find μ, σ. (6)
(c) Ten penguins are chosen at random. Find the probability at least four exceed 25 kg. (4)

a

b

c

Exercise 3E – Statistics Y2 pages 51-53 **Luisa · 39**

14. Adult dachshund lengths are normal. 20% are shorter than 16 inches and 10% exceed 18 inches.

(a) Find μ and σ. (6)

(b) Find the interquartile range. (2)

a

b

EXTENSION – Challenge P53 **Luisa · 40**

$X \sim N(\mu, \sigma^2)$ has interquartile range q.

(a) Express σ as a multiple of q, to 3 s.f.

(b) Explain why μ cannot be expressed in terms of q alone.

a

b

$X \sim B(n, p)$ counts successes in n trials, each with success probability p. What conditions are required?

The number of trials is fixed.

Each trial has two possible outcomes, success or failure.

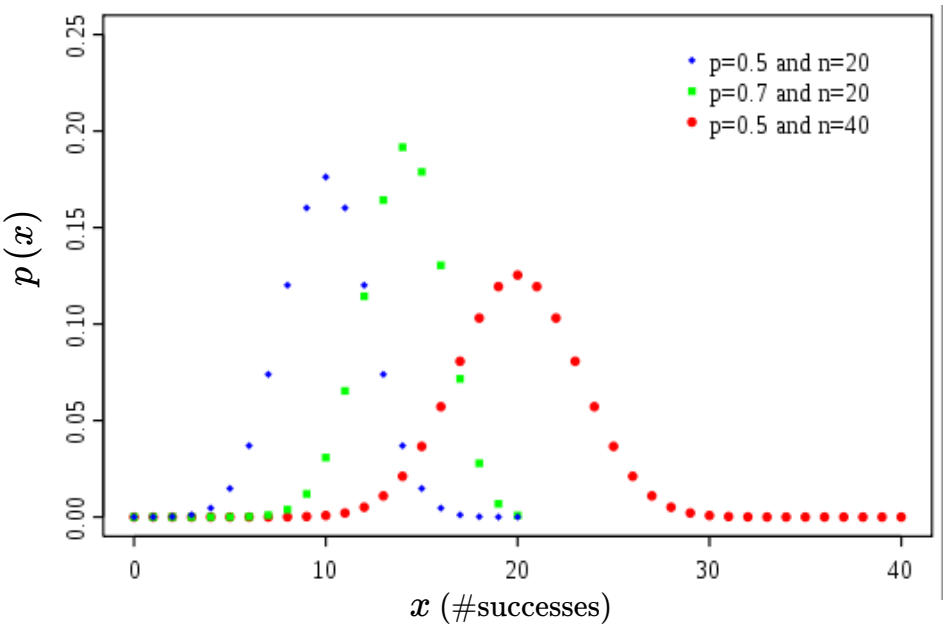
The probability of success is constant.

The trials are independent.

$P(X = r) = \binom{n}{r} p^r (1-p)^{n-r}$.

Examples of binomial counts

1. Flawed garments among 50 sampled.
2. Votes in favour of a new initiative among 3000 sampled voters, assuming independent decisions.
3. Left-handed students in a class of 34.

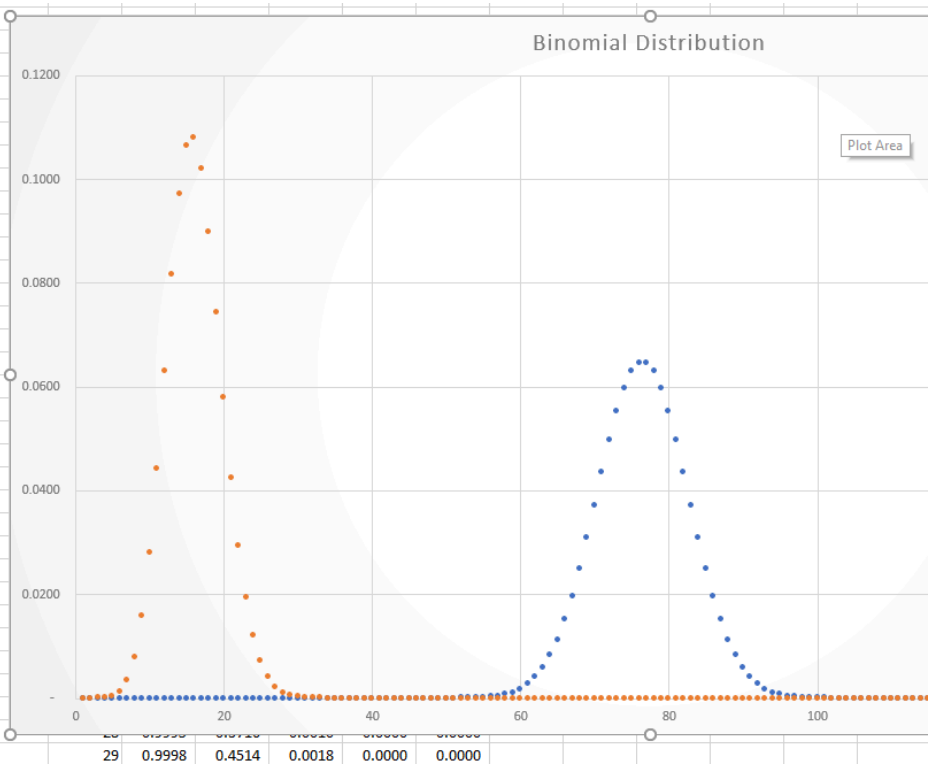


The graph shows several binomial probability functions. Which one resembles another distribution, and which distribution does it resemble?

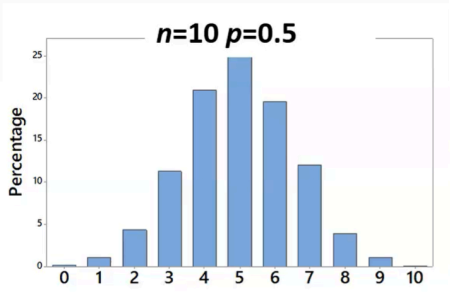
Background curiosity.. How to plot a binomial distribution in Excel

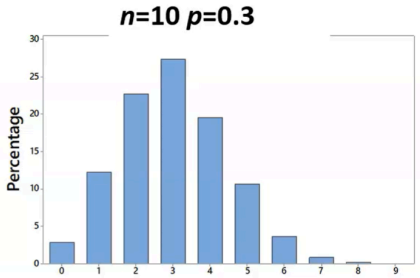
Microsoft Excel Worksheet

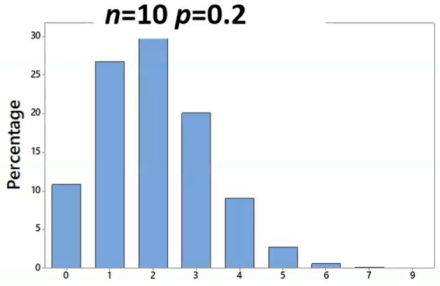
r	Normal 0.1	0.2	0.3	0.4	0.5
0	0.0000	0.0000	0.0000	0.0000	0.0000
1	0.0000	0.0000	0.0000	0.0000	0.0000
2	0.0000	0.0000	0.0000	0.0000	0.0000
3	0.0001	0.0000	0.0000	0.0000	0.0000
4	0.0004	0.0000	0.0000	0.0000	0.0000
5	0.0013	0.0000	0.0000	0.0000	0.0000
6	0.0035	0.0000	0.0000	0.0000	0.0000
7	0.0079	0.0000	0.0000	0.0000	0.0000
8	0.0159	0.0000	0.0000	0.0000	0.0000
9	0.0280	0.0000	0.0000	0.0000	0.0000
10	0.0443	0.0000	0.0000	0.0000	0.0000
11	0.0630	0.0000	0.0000	0.0000	0.0000
12	0.0817	0.0000	0.0000	0.0000	0.0000
13	0.0971	0.0001	0.0000	0.0000	0.0000
14	0.1063	0.0002	0.0000	0.0000	0.0000
15	0.1079	0.0004	0.0000	0.0000	0.0000
16	0.1019	0.0008	0.0000	0.0000	0.0000
17	0.0899	0.0016	0.0000	0.0000	0.0000
18	0.0744	0.0031	0.0000	0.0000	0.0000
19	0.0578	0.0054	0.0000	0.0000	0.0000
20	0.0424	0.0089	0.0000	0.0000	0.0000
21	0.0294	0.0138	0.0000	0.0000	0.0000
22	0.0193	0.0204	0.0000	0.0000	0.0000
23	0.0120	0.0286	0.0000	0.0000	0.0000
24	0.0071	0.0381	0.0000	0.0000	0.0000
25	0.0040	0.0484	0.0001	0.0000	0.0000
26	0.0022	0.0587	0.0001	0.0000	0.0000
27	0.0011	0.0679	0.0002	0.0000	0.0000
28	0.0005	0.0752	0.0005	0.0000	0.0000
29	0.0003	0.0797	0.0009	0.0000	0.0000
29	0.9998	0.4514	0.0018	0.0000	0.0000



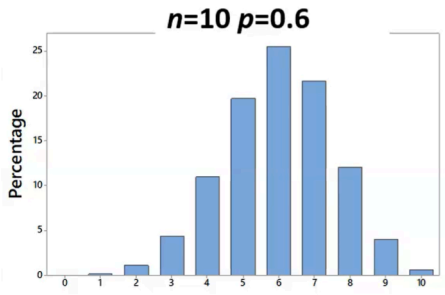
As p decreases the distribution becomes more positively skewed

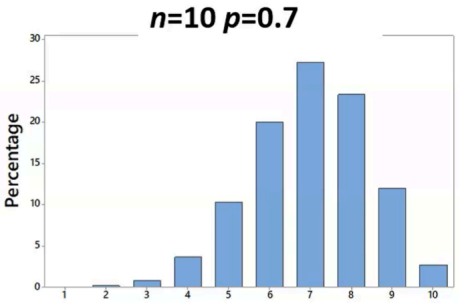


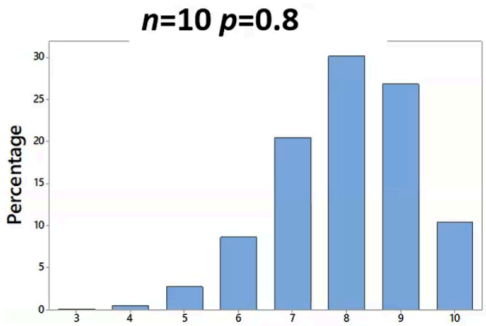




As p increases the distribution becomes more negatively skewed







If we're going to use a normal distribution to approximate a Binomial distribution, it makes sense that we set the mean and standard deviation of the normal distribution to match that of the original binomial distribution:

$\mu =$

$\sigma =$

If n is large and p close to 0.5, then the binomial distribution $X \sim B(n, p)$ can be approximated by the normal distribution $N(\mu, \sigma^2)$ where

$\mu = np$

$\sigma = \sqrt{np(1-p)}$

Quickfire Questions:

$X \sim B(10, 0.2) \rightarrow Y \sim$

$X \sim B(20, 0.5) \rightarrow Y \sim$

$X \sim B(6, 0.3) \rightarrow Y \sim$

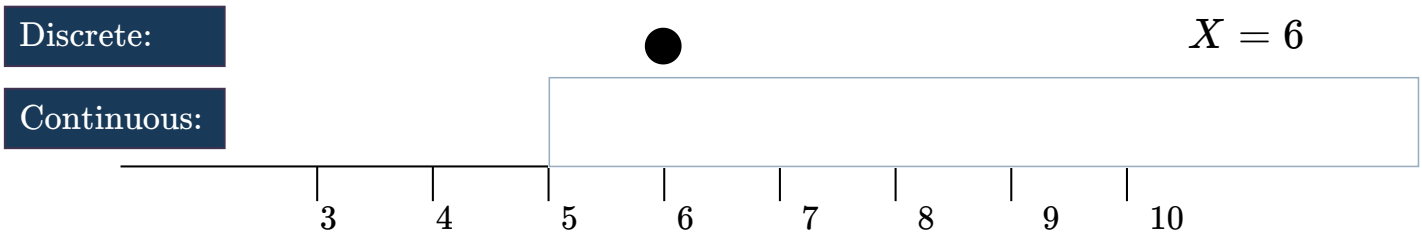
We tend to use the letter Y to represent the normal distribution approximation of the distribution X.

Why use a normal approximation?

- Tables for the binomial distribution only goes up to $n = 50$ and your calculator will reject large values of n.
- The formula for $P(X = x)$ makes use a factorials. Factorials of large numbers cannot be computed efficiently. Type in 65! for example; your calculator will hesitate! Now imagine how many factorials would be required if you wanted to find $P(X \leq 65)$.

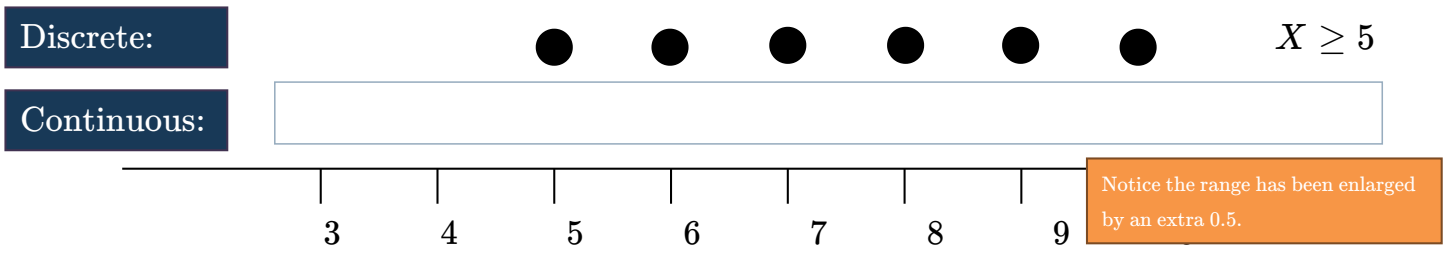
One problem is that the outcomes of a binomial distribution (i.e. number of successes) are **discrete** whereas the Normal distribution is **continuous**.
We apply something called a **continuity correction** to approximate a discrete distribution using a continuous one.

The random variable X represents the time to finish a race in hours. We're interested in knowing the probability Alice took 6 hours to the nearest hour.
How would you represent this time on a number line given hours is discrete? And what about if hours was now considered to be continuous (as Y)?

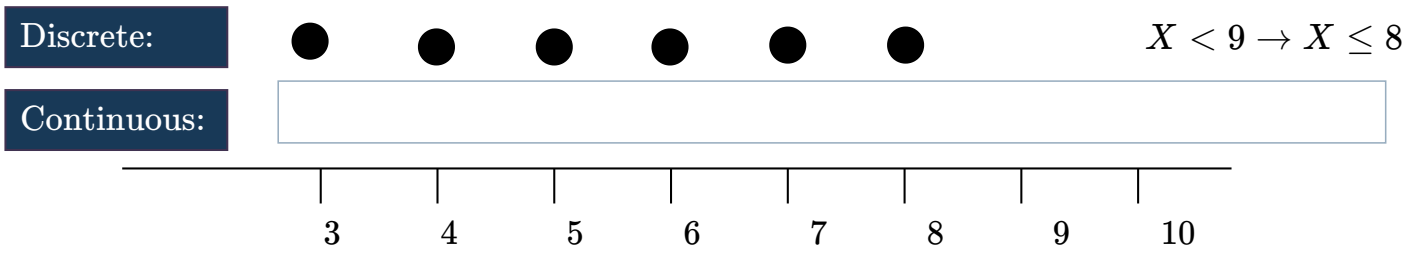


We can't just find $\mathbf{P}(\mathbf{Y} = \mathbf{6})$ when $\mathbf{Y}$ is continuous, because the probability is effectively 0. But $\mathbf{P}(\mathbf{5.5} < \mathbf{Y} < \mathbf{6.5})$ would seem a sensible interval to use because any time between 5.5 and 6.5 would have rounded to 6 hours were it discrete.

If X is a discrete variable, and Y is its continuous equivalent, how would you represent $P(X \geq 5)$ for Y?



How would represent $P(X < 9)$ for Y?



A continuity correction is approximating a discrete range using a continuous one.
1. If $>$ or $<$, convert to $\geq$, $\leq$ first.
2. Enlarge the range by 0.5.

Examples

Luisa · 49

Discrete probability	Continuous approximation
$P(X \leq 7)$	
$P(X < 10)$	
$P(X > 9)$	
$P(1 \leq X \leq 10)$	
$P(3 < X < 6)$	
$P(3 \leq X < 6)$	
$P(3 < X \leq 6)$	
$P(X = 3)$	

Continuity correction. First rewrite strict inequalities using integer endpoints and $\leq$ or $\geq$. Then extend the discrete range by 0.5 at each end.

Comparison on your calculators

Luisa · 50

400 children guess true or false on each of 80 quantum physics questions. What proportion of children get fewer than 38 correct?

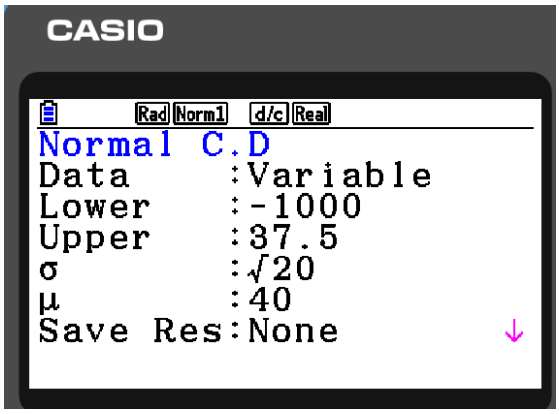
1. Find the normal distribution

2. Adjust for continuity

400 children guess true or false on each of 80 quantum physics questions. What proportion of children get fewer than 38 correct?

Normal Distribution

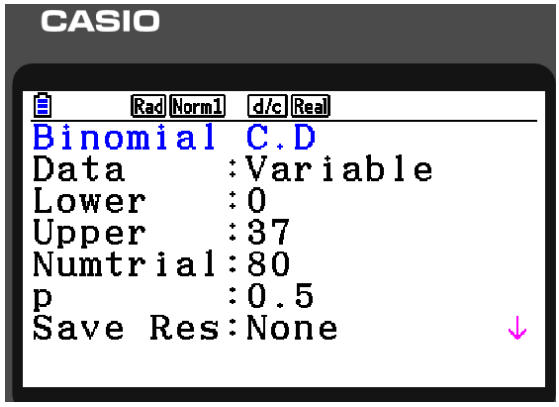
Write distribution



Display result

Binomial Distribution

Write distribution



Display result

[Textbook - Edited] For a particular type of flower bulbs, 55% will produce yellow flowers. A random sample of 80 bulbs is planted.
(a) Calculate the actual probability that there are exactly 50 flowers.
(b) Use a normal approximation to find a estimate that there are exactly 50 flowers.
(c) Hence determine the percentage error of the normal approximation for 50 flowers.

a a

b b

c c

Test Your Understanding

Edexcel S2 Jan 2004 Q3. $X \sim B(n, p)$.

(a) Which p gives the most accurate normal approximation? (1)

(b) Give a reason. (1)

(c) With $n = 200, p = 0.48$, find $P(90 \leq X < 105)$. (7)

a

b

c

Exercise 3F – Statistics Y2 pages 55

2. $X \sim B(150, 0.45)$. Use a suitable approximation to estimate:

(a) $P(X \leq 60)$ (b) $P(X > 75)$ (c) $P(65 \leq X \leq 80)$

2a 2b 2c

3. $X \sim B(200, 0.53)$. Use a suitable approximation to estimate:

(a) $P(X < 90)$ (b) $P(100 \leq X < 110)$ (c) $P(X = 105)$

3a 3b 3c

4. $X \sim B(100, 0.6)$. Use a suitable approximation to estimate:

(a) $P(X > 58)$ (b) $P(60 < X \leq 72)$ (c) $P(X = 70)$

4a 4b 4c

5. A fair coin is tossed 70 times. Estimate the probability of more than 45 heads using a suitable approximation.

5

6. A roulette ball lands on red with probability $\frac{50}{101}$. Estimate the probability of red in at least half of 1200 spins.

6

7. (a) Give two conditions for a normal approximation to the binomial. (2)

45% of orchids have pink flowers. In a random sample of 20, X are pink.

(b) Find $P(X = 10)$. (1)

A second random sample contains 240 orchids.

(c) Estimate the probability fewer than 110 are pink. (3)

(d) The probability at least q are pink is 0.2. Find q . (3)

a

b

c

d

8. 52% of drill bits last over 40 hours. Of 30 random bits X last over 40 hours.

(a) Find $P(X < 17)$. (1)

(b) Of 600 random bits, estimate the probability between 300 and 350 inclusive last over 40 hours. (3)

8a

8b

9. 56% of cereal boxes contain prizes. In a random sample of 100 boxes:

(a) Find the exact probability that 55 contain prizes. (1)

(b) Find the percentage error when a normal approximation is used. (4)

9a

9b

Distribution of Sample means — Luisa · 57

Choose three numbers independently and uniformly from the integers 1 to 100. One sample is 55, 74, 5. What is its mean?

Sample mean

What are the smallest and largest possible sample means?

Smallest

Largest

Which sample means are most likely? Which are least likely? What shape would the distribution have if we calculated the probability of every possible sample mean?

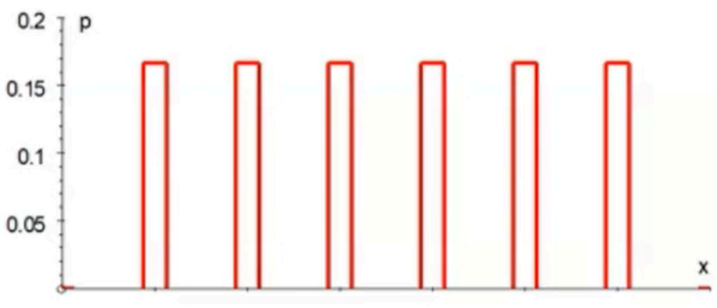
Roll of a fair die: sample size 1 — Luisa · 58

Roll one fair die. The population is $\{1, 2, 3, 4, 5, 6\}$. List the possible samples of size 1.

Samples

Probability distribution of the sample means

$\bar{x}$	Distribution
$P(\bar{X} = \bar{x})$	



Expectation

Variance

With a sample size of 1, the mean and variance equal those of the population.

Roll of a fair die: sample size 2 — Luisa · 59

Roll a fair die twice. How many ordered samples of size two are possible? What are their means?

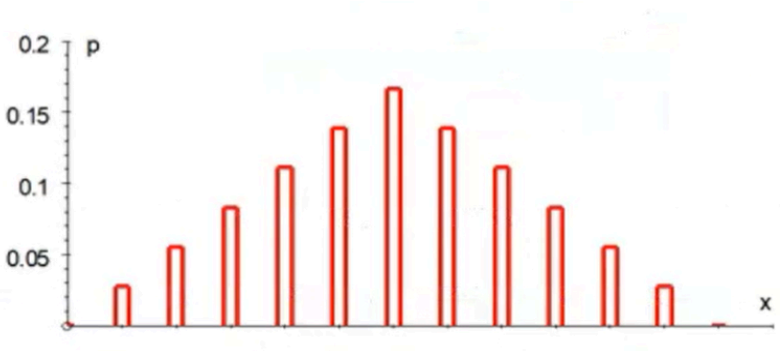
Sample means

Distribution

Roll of a fair die: sample size 2 — Luisa · 60

Distribution of the sample mean for two rolls of a fair die

$\bar{y}$	1	1.5	2	2.5	3	3.5	4	4.5	5	5.5	6
$P(\bar{Y} = \bar{y})$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$



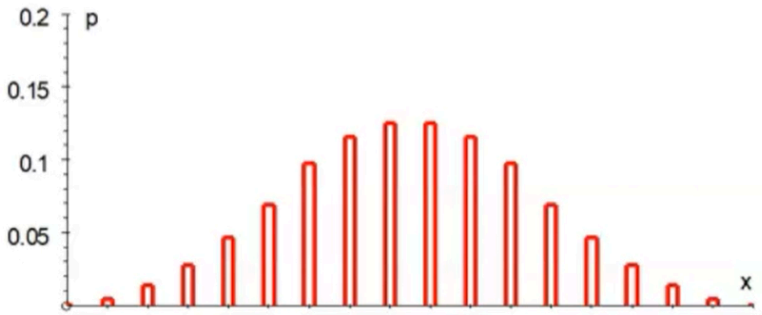
Expectation

Variance

Roll of a fair die: sample size 3 — Luisa · 61

Roll a fair die three times. Find the distribution of the sample mean, its expectation and its variance.

$\bar{z}$	Distribution
$P(\bar{Z} = \bar{z})$	



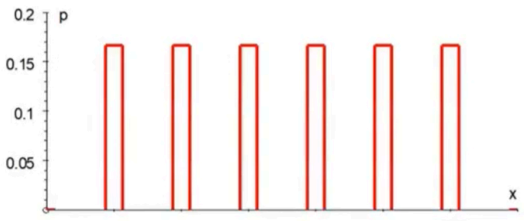
Expectation

Variance

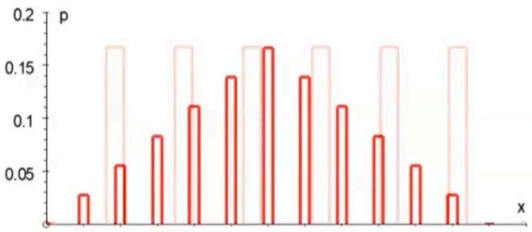
Pattern relating to the sample size — Luisa · 62

What changes as the sample size increases?

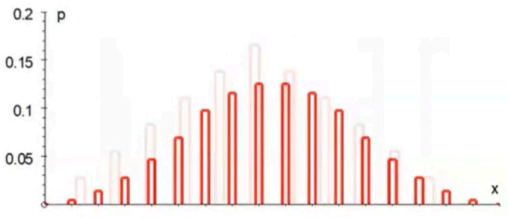
$n = 1$



$n = 2$



$n = 3$



As the sample size increases, the distribution of the sample mean becomes more bell-shaped.

How do mean and spread depend on sample size?

n	Mean	Variance	Standard deviation
1			
2			
3			

How do mean, variance and standard deviation change as sample size increases?

n	Mean	Variance	Standard deviation
1	3.5	$\frac{35}{12}$	$\sqrt{\frac{35}{12}}$
2	3.5	$\frac{35}{24}$	$\sqrt{\frac{35}{24}}$
3	3.5	$\frac{35}{36}$	$\sqrt{\frac{35}{36}}$

← **Sample size 1**
Mean $= \mu$, SD $= \sigma$

← **Sample size 2**
Mean $= \mu$, SD $= \sigma/\sqrt{2}$

← **Sample size 3**
Mean $= \mu$, SD $= \sigma/\sqrt{3}$

For independent observations with population mean μ and standard deviation σ:

1. The mean of the sample means is μ.
2. The standard deviation of sample means (the standard error) is $\sigma/\sqrt{n}$. It decreases as n increases.

For a random sample of size n from a normally distributed population X, the sample mean $\overline{X}$ is normally distributed with $\overline{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$

If the population is $N(\mu, \sigma^2)$, the mean of an independent random sample of size n also follows a normal distribution.

Individual observation

$X \sim N(\mu, \sigma^2)$

$z = \frac{x-\mu}{\sigma}$.

Sample mean

$\bar{X} \sim N(\mu, \sigma^2/n)$.

$z = \frac{\bar{x}-\mu}{\sigma/\sqrt{n}}$.

Calculator skills (Casio fx-CG50): Normal hypothesis testing

4 steps for setting up a Hypothesis Test for the population mean:

1. State $H_0 : \mu = m$, where m is the proposed population mean.

2. Choose the alternative: $\mu > m$ or $\mu < m$ for a one-tailed test, or $\mu \neq m$ for a two-tailed test.

3. Determine the significance level.

4. Consider the evidence from the sample.

Test results have mean 65 and standard deviation 10 and are normally distributed. After a new teacher arrives, eight students have mean score 72. Is there evidence of improvement at the 5% significance level?

The distribution of the sample mean is $\overline{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$

$$Z = \frac{\overline{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

Test scores are normal with mean 65 and standard deviation 10. After a new teacher arrives, eight students have a mean of 72. Test at 5% whether the population mean has increased.

In summary:

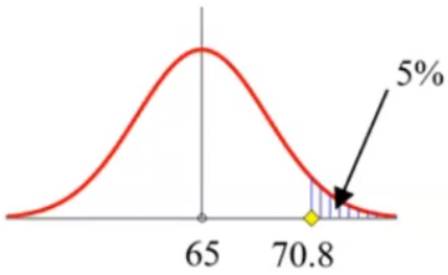
$H_0 : \mu = 65$, $H_1 : \mu > 65$, where μ is the population mean score.
Under H_0, $\bar{X} \sim N(65, 10^2/8)$ and $P(\bar{X} > 72) = 0.0239$.
$0.0239 < 0.05$, so reject H_0. There is evidence that the mean score has increased.

Method 2 Finding a critical region using a p-value — Luisa · 69

Test scores are normal with mean 65 and standard deviation 10. After a new teacher arrives, eight students have a mean of 72. Test at 5% whether the population mean has increased.

$$\bar{X} \sim N(65, 10^2/8)$$

Using your calculator you find that the critical region is $\bar{x} > 70.8$



The observed sample mean $\bar{x} = 72$ lies in the critical region. Reject H_0: there is evidence that the mean score has increased.

Example 1: Left Hand Tail using a p-Value — Luisa · 70

The supplier of LITE light bulbs claims a mean life of 130 hours. A trading standards organisation tests 400 bulbs after customer complaints. Their mean lifetime is 128.5 hours and the sample standard deviation is 13 hours. Test at the 2% level whether the population mean lifetime is lower than 130 hours.

Let X be the lifetime of a LITE bulb.

Mean

Variance

Hypotheses

p-value

Conclusion

Example 2: Left Hand Tail using a critical region **Luisa · 71**

The supplier of LITE light bulbs claims a mean life of 130 hours. A trading standards organisation tests 400 bulbs after customer complaints. Their mean lifetime is 128.5 hours and the sample standard deviation is 13 hours. Test at the 2% level whether the population mean lifetime is lower than 130 hours.

1. Let X be a bulb's lifetime. $H_0 : \mu = 130$, $H_1 : \mu < 130$.
Under H_0, $\bar{X} \approx N(130, 13^2/400)$.

2. Use inverse normal for the lower 2% tail.

Critical value

Critical region

Conclusion

Left Hand Tail: Method 2 using a critical region **Luisa · 72**

The supplier of LITE light bulbs claims a mean life of 130 hours. A trading standards organisation tests 400 bulbs after customer complaints. Their mean lifetime is 128.5 hours and the sample standard deviation is 13 hours. Test at the 2% level whether the population mean lifetime is lower than 130 hours.

1. Let X be a bulb's lifetime. $H_0 : \mu = 130$, $H_1 : \mu < 130$.
Under H_0, $\bar{X} \approx N(130, 13^2/400)$.

2. Use inverse normal for the lower 2% tail.

Critical value

Critical region

Conclusion

Two tailed tests

In a one-tailed test the alternative specifies a direction: $\mu < k$ or $\mu > k$.

In a two-tailed test $H_1 : \mu \neq k$. A difference in either direction is relevant.

There are two parts to the critical region (both tail of the distribution are considered)

- At the 5% significant level the critical region will have 2.5% in each tail
- When considering the p values compare the value with half of the significance level as you are just comparing with one of the tails

Two tailed test example

Leaves of a rare plant species have normal lengths with mean 8.6 cm and standard deviation 1.2 cm. A botanist finds a clump of plants and measures 50 leaves. Their lengths total 442 cm. Test at the 5% level whether the mean differs from 8.6 cm. What should the botanist conclude?

Let X be leaf length. Under H_0, $\bar{X} \sim N(8.6, 1.2^2/50)$. In general the sample mean has variance σ^2/n.

$H_0 : \mu = 8.6, H_1 : \mu \neq 8.6.$

Use the sample mean of the 50 measured leaves.

$\bar{x} = 442/50 = 8.84$ cm.

$P(\bar{X} > 8.84) = 1 - 0.92135\ldots = 0.0787.$

$0.0787 > 0.025$, so do not reject H_0. The results do not establish that the plants are of a different species.

The 97.5th percentile is 8.93 cm. The upper rejection region is $\bar{X} > 8.93$. Our mean of 8.84 cm lies inside the acceptance interval.

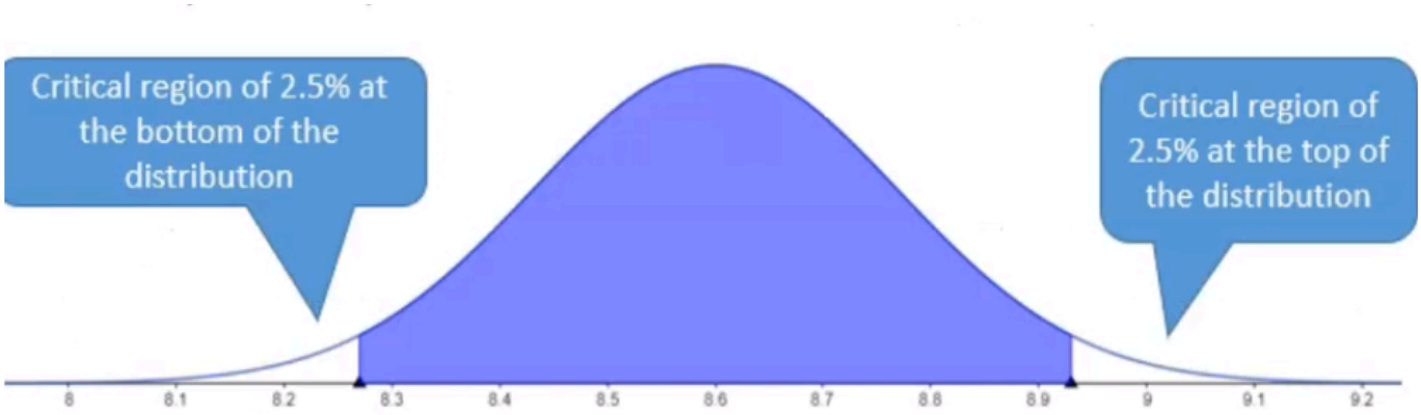
Two tailed test example

Luisa · 75

Leaves of a rare plant species have normal lengths with mean 8.6 cm and standard deviation 1.2 cm. A botanist finds a clump of plants and measures 50 leaves. Their lengths total 442 cm. Test at the 5% level whether the mean differs from 8.6 cm. What should the botanist conclude?

$\bar{X} \sim N(8.6, 1.2^2/50)$.

$H_0 : \mu = 8.6$, $H_1 : \mu \neq 8.6$. The critical region has 2.5% in each tail.



Exercise 1

Luisa · 76

1. A random sample of size n is taken from a normal population with known standard deviation σ. Find the p-value and carry out each hypothesis test at the indicated significance level.

	σ	n	$\bar{x}$	H_0	H_1	Level
(i)	14	20	247	$\mu = 240$	$\mu \neq 240$	2%
(ii)	15	10	172.9	$\mu = 165$	$\mu > 165$	5%
(iii)	24	15	226	$\mu = 240$	$\mu < 240$	1%

(i)

(ii)

(iii)

Exercise 2

Luisa · 77

Adult student masses are normal with mean 67.4 kg and standard deviation 3.8 kg. A sample of 24 has mean 65.8 kg. Assuming unchanged standard deviation, test at the 1% level whether the mean has decreased. Give the critical region.

2

Exercise 3

Luisa · 78

A machine produces ball bearings of intended diameter 15 mm. A random sample of 80 has mean 14.64 mm. Diameters are normal with known standard deviation 1.41 mm. Test at the 2% significance level whether the mean has changed, giving the critical region.

3

Textbook Examples — Luisa · 79

Fruit juice carton volumes are normally distributed with standard deviation 3 ml. The company claims a mean of 60 ml. An inspector suspects that it is overstated. A random sample of 16 cartons has mean 59.1 ml. State the hypotheses and test the complaint at the 5% level.

Null hypothesis

Alternative hypothesis

Variable

Population

Sample mean

Test and conclusion

One carton and a sample mean X describes an individual carton. $\bar{X}$ describes the mean of a random sample of 16 cartons. Start from the distribution of X under H_0, then divide its variance by 16 to get the distribution of $\bar{X}$.

Finding the critical region — Luisa · 80

A machine produces bolts of normally distributed diameter, with mean 0.580 cm and standard deviation 0.015 cm. After servicing, a random sample of 50 bolts is taken. The standard deviation is unchanged.
(a) State the hypotheses and find the 1% critical region for a change in mean diameter.
(b) The sample mean is 0.587 cm. Comment.

Hypotheses

Population

Sample mean

Critical values

Critical region

Observation

Two tails at 1% give 0.5% in each tail. Use the normal table or inverse normal on your calculator.

p	z	p	z
0.5000	0.0000	0.0500	1.6449
0.4000	0.2533	0.0250	1.9600
0.3000	0.5244	0.0100	2.3263
0.2000	0.8416	0.0050	2.5758
0.1500	1.0364	0.0010	3.0902
0.1000	1.2816	0.0005	3.2905

Standardise using the sample mean's standard error $\sigma/\sqrt{n}$.

Test Your Understanding — Luisa · 81

Edexcel S3 June 2011 Q7(a). Roastie's Coffee packets claim to weigh 250 g. A supermarket manager suspects the actual mean is lower. A random sample of 90 has mean 248 g and standard deviation 5.4 g.

(a) Test the manager's claim at the 5% level, stating the hypotheses. (5)

a

Exercise 3G – Statistics Y2 pages 58-60 — Luisa · 82

6. Bolt diameters D are normal with standard deviation 0.1 mm; 5% exceed 5.62 mm. Only diameters from 5.1 to 5.6 mm can be sold.

(a) Find the probability a bolt can be sold. (5)

(b) Of 12 bolts, find the probability fewer than three cannot be sold. (2)

(c) A second machine makes normally distributed diameters with standard deviation 0.08 mm. The mean of 20 is 5.52 mm. Test whether the population mean is less than 5.7 mm, at 2.5%. State the hypotheses. (4)

a

b

c

7. European water vole masses M are normal with standard deviation 12 g. 2.5% exceed 160 g.

(a) Find the mean. (3)

(b) Of eight random voles, find the probability at least four exceed 150 g. (3)

(c) Water rat masses are normal with standard deviation 85 g. For a random sample of 15 the mean is 875 g. Test whether the population mean differs from 860 g at 10%. State the hypotheses. (4)

a

b

c

8. Daily mean windspeeds are normal with standard deviation 3.1 knots. A random sample of 25 days at Heathrow in 2015 has mean 12.2 knots. Test at the 2.5% significance level whether the population mean exceeds 9.5 knots. State the hypotheses. (4)

8

Conditional Probabilities — Luisa · 85

Edexcel S1 May 2014(R) Q4(c). The flight time from London to Malaga is normally distributed with mean 150 minutes and standard deviation 10 minutes. The flight time X minutes from London to another city is normally distributed with mean μ.

Given $P(X < \mu - 15) = 0.35$, find $P(X > \mu + 15 \mid X > \mu - 15)$. (3)

(c)

Test Your Understanding — Luisa · 86

Edexcel S1 Jan 2013 Q4(a,c). The time L hours that a phone lasts before it needs charging is normally distributed with mean 100 hours and standard deviation 15 hours.

(a) Find $P(L > 127)$. (3)

(b) Alice is about to make a six-hour journey. It has been 127 hours since she charged the phone. Find the probability it lasts to the end of her journey. (4; original part c)

a

b

IQ is distributed as $X \sim N\left(100, 15^2\right)$.
A person is considered to be 'pretty smart' if their IQ is at least 130.
Determine the median IQ of 'pretty smart' people.